## Section A



**1** **(a)** $F \propto Mm / R^2$ ……..(*words or explained symbols*) ............................................M1
*either* $M$ and $m$ are point masses
*or* $R >>$ diameter of masses …(*do not allow 'size'*) ...................................... A1 [2]

**(b)** **(i)** equatorial orbit ................................................................................................ B1
period 24 hours / same angular speed ...................................................................... B1
from west to east / same direction of rotation ....................................................... B1 [3]
(*allow one of the last two marks for 'always overhead' if 2nd or 3rd marks not scored*)

**(ii)** gravitational force provides centripetal force
/ gives rise to centripetal acceleration ….(*in 'words'*) ........................................ B1
$GM / x^2 = x\omega^2$ ..........................................................................................M1
$g = GM / R^2$ ...............................................................................................M1
to give $gR^2 = x^3\omega^2$ ................................................................................... A0 [3]

**(iii)** $\omega = 2\pi / (24 \times 3600) = 7.27 \times 10^{-5}$ rad s$^{-1}$ ................................................C1
$9.81 \times (6.4 \times 10^6)^2 = x^3 \times (7.27 \times 10^{-5})^2$ ..................................................C1
$x^3 = 7.6 \times 10^{22}$
$x = 4.2 \times 10^7$ m ........................................................................................ A1 [3]
(*use of $g$ = 10 m s$^{-2}$, loses 1 mark but once only in the Paper*)

**[Total: 11]**

**2** **(a)** *either* $pV = NkT$ *or* $pV = nRT$ *and* $n = N / N_A$ .................................... C1